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Neural and Cognitive Correlates of Adolescent Anxiety: Widespread Brain Network Alterations, Increased Control Costs, and Visuospatial Processing Deficits --Manuscript Draft--

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Keywords:	Anxiety; Adolescence; Brain development; Functional networks; Brain network controllability; cognitive function

Neural and Cognitive Correlates of Adolescent Anxiety: Widespread Brain Network Alterations, Increased Control Costs, and Visuospatial Processing Deficits

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Methods: The study included $n=3499$ youth (median(IQR) age =12.0(1.1) years; ~51.0% females) from the Adolescent Brain Cognitive Study. Resting-state fMRI, structural MRI, anxiety, and neurocognitive data were analyzed. Mixed effects models investigated associations between anxiety, brain characteristics (topological and morphometric properties and costs of network control), and cognitive outcomes.

Results: Females and older youth and higher anxiety ($\beta=0.04$ -1.15, 95%CI=[0.01,2.18]). Increased connectivity/centrality of the cerebellum, amygdala, and thalamus, and higher cerebellum-basal-ganglia-thalamus control costs were associated with higher anxiety ($\beta=0.05$ -0.06, 95%CI=[0.01,0.11]). In contrast, temporoparietal and frontoparietal network efficiency and resilience were protective for social and/or generalized anxiety disorders (adjusted odds ratio (aOR)=0.72-0.80, 95%CI=[0.62,0.98]). Higher cortical thickness/volume of the superior temporal gyrus, pars opercularis, precentral gyrus, and putamen were associated with higher odds of generalized or social anxiety disorder (aOR=1.28-1.44, 95%CI=[1.07,1.78]). Finally, anxiety was associated with lower response accuracy in the Little Man Task ($\beta=-0.05$, 95%CI=[-0.09,-0.01]), and inversely moderated brain-task performance associations.

Conclusions: Adolescent anxiety has widespread brain correlates, including higher cost of controlling the cerebellum-basal ganglia-thalamus network, which plays a key

role in emotion regulation. It also correlates with visuospatial processing deficits, both directly and through its neural substrates.

KEY POINTS

- Adolescent anxiety is associated with alterations in brain structures and circuits that play key regulatory roles in emotional processing, social function and decision-making
- Intrinsic dynamics of the cerebellar-basal ganglia-thalamic network, which plays a central role in emotional regulation, is harder to control in youth with anxiety
- Social anxiety is associated with structural and topological alterations of the temporoparietal and frontoparietal networks, which mature significantly in adolescence
- Anxiety is directly and indirectly (through its neural correlates) associated with deficits in visuospatial processing

1. INTRODUCTION

Adolescence and young adulthood are sensitive periods for mental health. Common disorders often emerge in adolescence (Paus et al., 2008), a period of profound biological changes and complex interactions between genetic, environmental and experiential factors (Powers et al., 1989; Kessler et al., 2005; Schlack et al., 2021; Solmi et al., 2022). In particular, anxiety, a family of disorders that encompasses generalized anxiety (GAD), social anxiety (SAD), separation anxiety, panic disorder, and various phobias, affects over 20% of adolescents in the US (Bandelow and Michaelis, 2015; CDC, 2025), and its prevalence increases with age (Cohen et al., 1993; Fergusson et al., 1993; Costello et al., 1996, 2006). In addition to genetic factors, social stressors, including poor relationships with parents and peers, negative family dynamics, and physical environmental factors, such as noise, pollution, and unsafe neighborhoods, increase the risk of anxiety (Robinson et al., 1992; Stark et al., 1993; Bögels and Brechman-Toussaint, 2006; Rapee, 2012; Teo et al., 2013; Waszczuk et al., 2014; Cheesman et al., 2018; Lawrence et al., 2019; Wang et al., 2020; Elmore and Crouch, 2020; Crandon et al., 2022; Buttazzoni et al., 2022; Lou et al., 2025). Similarly, negative experiences, such as trauma, abuse and neglect are also significant risk factors (Nanda et al., 2016; Lee et al., 2020; Chen and Qin, 2020; Colich and McLaughlin, 2022; Huang et al., 2023; Marusak et al., 2023; Marini et al., 2024; Wang et al., 2024; Calaresi et al., 2025).

Anxiety can have detrimental effects on cognitive function, including attention, memory, reward and emotional processing, decision-making and social function (Tallon et al., 2016; Maloney et al., 2014; Grupe, 2017; Kilford et al., 2023; Loreto and Joao, 2025). In adolescents, higher anxiety levels are associated with impaired emotional processing, including higher bias to threat-related inputs, poor academic performance deficits in working memory, and risk aversion (Muris and Field, 2008; Owens et al., 2012; Dudeney et al., 2015; Ducay and Alava, 2021; Manuel et al., 2023; Wolters et al., 2023; Rabner et al., 2024). Although the mechanisms through which anxiety adversely impacts cognitive function are incompletely understood, prior adult studies have reported widespread circuit-level alterations underlying cognitive problems in individuals with anxiety. These include abnormally stronger connections in the salience network, and stronger connections between the amygdala and the cingulo-opercular network, the striatum, default and salience networks, and the cerebellum (Milad et al., 2007; Marchand, 2010; Xu et al., 2021; Jackson and Bernard, 2022; Zharova et al., 2025; Langhammer et al., 2025). Pediatric studies in have reported abnormally stronger connections between and within limbic regions (especially the amygdala) (Perino et al., 2021; Sievertsen et al., 2025), and the salience and somatomotor networks (Lee et al., 2021), and stronger connections between the amygdala, the basal ganglia (especially the striatum) and the cerebellum in youth with GAD and SAD (Roy et al., 2013). Aberrant alterations in the cerebellum's morphology, connectivity and projections to other brain areas have also been reported (Chin et al., 2025), a structure that plays a key regulatory role in emotional processing and inhibitory control through its

connections with the amygdala, the thalamus and the prefrontal cortex. Similarly, abnormally stronger connections between the thalamus, the insula and amygdala, but weaker connections between the thalamus and the PFC have also been reported (Janiri et al., 2020; Chun et al., 2025).

Beyond connectivity, anxiety has also been associated with structural alterations, some of which are common to children and adults with the condition, while others are distinct in the two populations. Lower gray matter volume of the anterior cingulate and insula, ventromedial and dorsolateral prefrontal cortex (PFC), superior and inferior temporal gyrus, the thalamus, the caudate and the putamen has been reported in adults with anxiety disorders (Moon et al., 2014; Chen et al., 2020; Liu et al., 2024), whereas higher gray matter volume of the superior temporal gyrus and the thalamus, higher cortical thickness in the ventrolateral and ventromedial PFC and the precentral gyrus, and lower hippocampal and amygdalar volume have been reported in adolescents (De Bellis et al., 2002; Liao et al., 2013; Strawn et al., 2013, 2014; Gold et al., 2017; Hammoud et al., 2024).

Together, prior studies have identified complex, potentially bidirectional associations between anxiety and distributed brain structures and circuits, especially the amygdala, basal ganglia, thalamus, prefrontal cortical areas, and the cerebellum, which are part of networks that support extensive cognitive and regulatory processes and mental health, and undergo significant changes during periods of heightened maturation, such as

adolescence. However, associations between anxiety, these distributed networks, and cognitive processing have not been well characterized in youth, an important gap in knowledge and barrier to the development of early interventions to improve cognitive and mental health outcomes in youth with anxiety disorders. This study aimed to address this critical gap, leveraging large-scale neuroimaging, neurocognitive and mental health data from a large adolescent sample in Adolescent Brain Cognitive Development (ABCD) study (Casey et al., 2018) that captured the heterogeneity of neural and cognitive development.

Recent studies using data from the ABCD cohort have provided important insights into individual, environmental and experiential correlates of adolescent anxiety. They showed that females, youth in later pubertal stages, from low-income families, and those who spent longer times on social media and/or using screens had higher anxiety and risk of developing related disorders (Assari, 2020; Keyin et al., 2025; Lima et al., 2025; Duan et al., 2025; Parker et al., 2026). Also, adolescents with anxiety had abnormally stronger connections between limbic regions and elements of the default, salience, frontoparietal control and dorsal attention networks (Thomas et al., 2024; Sievertsen et al., 2025). Those with lower amygdala, hippocampal, and precentral gyrus volumes had higher odds of an anxiety disorder (Hammoud et al., 2024). Finally, youth with SAD had abnormally stronger connections between the default, salience, cingulo-opercular and retrosplenial-temporal networks (Hickson et al., 2026). Also, a recent study showed that brain connectivity is predictive of anxiety and depression in

adolescence (Morfini et al, 2025). Despite their valuable insights, these studies have investigated morphometric or characteristics separately, and have not examined other aspects of brain function, such as topological circuit organization beyond connectivity and intrinsic network dynamics, which play key roles in cognitive processing. Finally, most of these studies have not provided mechanistic insights into cognitive function in youth with anxiety by linking brain abnormalities to cognitive deficits.

In a sample of ~3500 youth from the ABCD two-year follow-up cohort, we investigated the neural and cognitive correlates of anxiety, and associations between brain abnormalities and cognitive performance across domains. We hypothesized that anxiety is associated with aberrant topological organization of developing brain circuits beyond the strength of their connections, especially those supporting emotional processing and social function, c) morphometric alterations in these structures, and d) atypical task-independent (intrinsic) dynamics, which, in turn, adversely impact cognitive processing across domains. Our and others' work has shown that temporal fluctuations of spontaneous brain activity and network coordination are developmental markers (Faghiri et al., 2018; Sydnor et al., 2023; Lim et al., 2025) and have shown that psychiatric and developmental disorders are associated with aberrant dynamic functional connectivity (de Lacy and Calhoun, 2018; Fu et al., 2021). However, associations between anxiety and temporal fluctuations of topological patterns and spontaneous activity in adolescents have not been previously characterized. In addition, intrinsic dynamics are regulated by endogenous (e.g., biochemical) processes that are

not directly measurable in the intact human brain. In prior work, we have used a closed-loop control theoretic framework to model the feedback loop between these processes and the brain, and have shown that intrinsic dynamics are regulated (controlled) by a sparse set of cognitive and topological hubs, i.e., highly connected regions that integrate distributed, domain-specific computations in response to cognitive demands (Lim et al., 2025). In the present study, we hypothesized that anxiety is associated with changes in the brain's controllability as well, and thus 'costs' of regulating its dynamics, as a result of topological and structural alterations. Finally, we hypothesized that through its impact on structure, function and dynamics of underdeveloped brain networks, anxiety is negatively associated with cognitive processing in domains that continue to evolve in adolescence, including memory, emotional processing, decision-making, and overall cognitive flexibility.

2. METHODS and MATERIALS

The study was approved by the Boston Children Hospital Institutional Review Board. All analyzed data are publicly available through the NIH Brain Development Cohorts (NBDC) Data Sharing Platform (NBDC, 2025).

2.1 Participants

N = 3499 youth (median (IQR) age = 12.0 (1.1) years; 50.9% females) from the ABCD two-year follow-up assessment were included in the study, based on the following criteria: a) availability of both anxiety and neuroimaging data (structural MRI and at least

one 5-min resting-state fMRI run) of adequate quality for analysis (less than 10% frames censored for motion based on a displacement threshold of 0.3 mm), b) no clinical MRI findings, c) no diagnosis of a developmental or neuropsychiatric disorder, since these disorders adversely affect brain structural characteristics and circuit organization in disorder-specific ways (Cherkassky et al., 2006; Assaf et al., 2010; Konrad and Eickhoff, 2010; Müller et al., 2011; Chase and Phillips, 2016).

2.2 Data

2.2.1 Measures of Anxiety

Anxiety raw and t-scores (standardized for age and sex) from Child Behaviour Checklist (CBCL) were analyzed (Wadsworth et al., 2001). Some ABCD studies on youth anxiety have analyzed raw (Cheng et al., 2021; Sisk et al., 2022; Vargas and Mittal, 2023; Roberts et al., 2023; Estévez-López et al., 2023; Parker et al., 2025; Shi et al., 2025) and others t-scores (Menon and Krishnamurthy, 2021; Serio et al., 2022; Correa et al., 2023; Marusak et al., 2023; Sievertsen et al., 2025; Johnson et al., 2025; Ghaleb et al., 2025; Sheldrake et al., 2025; Parker et al., 2026). Individual anxiety disorders, including generalized anxiety disorder (GAD), social anxiety disorder (SAD) and separation anxiety, were also analyzed and are represented by binary variables in the ABCD datasets (Hammoud et al., 2024). Finally, another binary anxiety variable was created to represent any type of anxiety, i.e., 1 = any diagnosed anxiety disorder or at least subclinical scores ($t \geq 65$), and 0 otherwise.

2.2.2 Imaging Modalities

Morphometric characteristics that had been estimated by the ABCD from fully preprocessed structural MRI, and minimally preprocessed resting-state fMRI (rs-fMRI), which had undergone initial preprocessing at the Data Analysis, Informatics & Resource Center (DAIRC) (Hagler et al, 2019) were analyzed. These data were acquired across the 21 ABCD sites, using 3.0T Siemens, GE or Phillips scanners (repetition time (TR) = 0.8 s; resolution = 2.4 mm isotropic). Minimally preprocessed rs-fMRI data (to correct for B0 distortion and assess quality based on controls set by the ABCD) were further processed by the custom Next Generation Neural Data Analysis (NGNDA) pipeline (NGNDA, 2020), to coregister the fMRI to the participant's structural fMRI, normalize to a common atlas, correct for motion-related artifacts and censor frames based on a 0.3 mm displacement threshold, suppress cardiorespiratory and other artifacts, and harmonize the data through amplitude normalization, to account for signal differences between scanners and head coils (32- vs 64-channels) (Brooks et al, 2021). Voxel fMRI signals were then downsampled to parcel signals (taking the median of voxels within a parcel), to facilitate computationally tractable analyses. Three atlases were used for this purpose, the Schaefer cortical atlas (1000 parcels) (Schaefer et al., 2018), the Melbourne subcortical atlas (54 parcels) (Tian et al., 2020) and the probabilistic Diedrichsen cerebellar atlas (34 parcels) (Diedrichsen et al., 2009). Topological network properties were estimated at this resolution. Similarly to our prior work (Lim et al, 2025a, b), in order to conduct tractable estimations of control costs, intrinsic dynamics and information transfer, the data were further downsampled at a resolution of 100 regions,

using the parcellation of resting-state networks in Yeo et al. (2011), to group parcels into larger regions. In addition to one high-quality rs-fMRI run (based on lowest percent of frames censored for motion (median (IQR) = 1.1 (4.0) %) and lowest median connectivity (assuming that the brain at rest is overall weakly synchronized), 2687 participants (76.6%) had a second rs-fMRI run of adequate quality for analysis (median (IQR) = 1.6 (4.3) percent of frames censored for motion).

2.3 Topological analysis of time-compressed networks and regions

The peak signal cross-correlation was used as the estimator of pairwise connection strength. Cohort-wide statistical thresholds were estimated from connectivity matrices across all brains and fMRI runs with less than 10% of frames censored for motion (Brooks et al., 2021). The upper confidence interval of the peak cross-correlation's moderate outlying value (median + 1.5*IQR) was selected as a conservative threshold. Adjacency matrices were estimated from connectivity matrices using this threshold. Topological properties were estimated from these matrices at two spatial scales: individual regions and large-scale networks. In addition to the 17 resting-state networks delineated in (Yeo et al., 2011), the reward, prefrontal, social and central executive networks were analyzed (Blakemore, 2008; Haber and Knutson, 2010; Sherman et al., 2014). Additional circuits that combined regions implicated in anxiety in prior studies, including the amygdalo-thalamic, thalamo-basal ganglia, and amygdalo-basal ganglia circuits, were also constructed, as well as a larger network encompassing the basal ganglia (striatum, caudate, putamen, nucleus accumbens), thalamus (ventral anterior

and posterior, dorsal anterior and posterior), cerebellum and amygdala (a total of 88 parcels). Network-level properties and node-level properties were estimated using The Brain Connectivity Toolbox (Rubinov and Sporns, 2010) and additional algorithms in the NGNDA pipeline. They included connectivity within and across networks, efficiency, modularity, topological stability, robustness and fragility, local connectedness (node degree), local clustering and regional importance in the network (eigenvector centrality) (Wu et al., 2011; Brooks et al., 2021; Pasqualetti et al., 2020).

2.4 Analysis of intrinsic network dynamics and controllability

2.4.1 Intrinsic brain networks and temporal fluctuations of topological properties

Time-varying connectivity matrices were estimated from fMRI signals using two approaches: a) using a sliding window (20 frames (16.0 s) long, advanced by one frame) a covariance matrix was estimated, and was then transformed to a correlation matrix (the absolute correlation matrix was used for estimating time-varying topological properties) (Lim et al., 2025); b) each signal was transformed to its complex counterpart using the Hilbert Transform, and the instantaneous phase difference between pairs of signals was estimated. The absolute cosine of this difference was used as a measure of pairwise correlation. This estimation was done at the resolution of 1088 parcels, and resulting connectivity matrices were then downsampled to the resolution of 100 regions. Brain-specific connectivity thresholds were then estimated as in our prior work (Lim et al., 2025), and were used to estimate time-dependent weighted adjacency matrices. In turn, these (and corresponding binary matrices) were used to

estimate time-dependent topological properties. The coefficient of dispersion (ratio of the interquartile range over the sum of the first and the third quartiles) was used as the estimator of these properties' temporal variability).

2.4.2 Controllability of intrinsic dynamics

A detailed description of the closed-loop, sparsity-promoting control framework used in the present study is described in Mitrai et al. (2023) and Lim et al. (2025) . Briefly, the brain is assumed to be a linear dynamic system that is regulated by a controller that balances cost of exerting its control action with spatial sparsity of that action, i.e., the controller only needs to exert its action on a sparse set of brain regions in order to regulate the entire network. This framework was used to estimate feedback costs associated with each brain region and identify regions that maintain their control action over the network even at high costs. This analysis was conducted at the spatial resolution of 100 regions. In addition to region-specific feedback costs, median costs over each network of interest were also estimated.

2.4.3 Temporal fluctuations of intrinsic BOLD activity

Amplitude of intrinsic signal fluctuations was estimated from each parcel rs-fMRI time series spectrum. The square root of the spectrum in the frequency range 0.01 - 0.1 Hz was used as a measure of fluctuation amplitude (Sydnor et al., 2023; Lim et al., 2025). Parcel-level estimates were then downsampled to 100 regional estimates for consistency with other analyses at this resolution.

2.5 Morphometric properties

These properties were estimated by the ABCD from T1-weighted MRI (Hagler et al, 2019), parcellated using the Desikan-Killiany cortical atlas (68 regions) (Desikan et al, 2006) and a probabilistic subcortical atlas (30 regions) (Fischl et al., 2002). They included cortical thickness, grey matter volume (cortical and subcortical), white matter volume (cortical) and cortical and subcortical white matter intensity.

2.6 Neurocognitive data

The ABCD study conducts extensive neurocognitive assessments using primarily the NIH Toolbox Cognition Battery (Luciana et al., 2018), with additional tasks, including the Little Man task, Emotional Stroop task, Delay Discounting task and Game of Dice tasks (Acker, W and Acker, C, 1982; Smolker et al., 2022; Gelino et al., 2025; Thompson et al, 2019). Although associations between anxiety, brain measures and cognitive performance were examined across tasks for which data were available at the two-year follow-up, the present study focused on domains that continue to evolve in adolescence, including inhibitory control, executive function, decision-making and risk assessment, cognitive flexibility, language, and memory, assessed with the: a) Flanker Inhibitory Control task (fully corrected for demographics t-scores), b) Rey Auditory Verbal Learning task (RAVLT; raw scores for total correct responses), c) Oral Reading Recognition task and d) Peabody Picture Vocabulary task (fully corrected t-scores for both tasks), e) Game of Dice Task (raw net score), f) monetary Delayed Discounting

task (area under the indifference points curve), g) Little Man Task (number of correct responses and accuracy), and Emotional Stroop task (accuracy measured by the proportion of correct responses to congruent and incongruent trials, and reaction time).

2.7 Statistical Analysis

Generalized linear and logistic mixed effects models were used to analyze the data, and account for ABCD site effects through the inclusion of a random intercept and slope for site. Linear models tested associations between anxiety scores (the dependent variable) and brain measures (and similarly for demographic characteristics), or between cognitive outcomes (the dependent variables), anxiety and/or brain measures. Logistic models tested associations between individual anxiety disorders (or the binary anxiety variable) and brain and/or demographic parameters. In these models, the independent variable of interest (for example brain measures) was standardized as a z-score, for better interpretation of reported adjusted odds ratios (aOR). Additional moderation models were developed to test the moderating impact of anxiety on associations between brain measures and cognitive outcomes. These models included additive brain and anxiety parameters and an interaction term for their product.

All analyses were adjusted for sampling differences between sites using the propensity scores provided by the ABCD. All models included age, sex (male = 1, female = 2), ethnicity, race (as indicator variables), BMI z-score (Brooks et al., 2023), pubertal stage (an ordinal variable), estimated from the parent-reported Pubertal Development scale

(Petersen et al., 1988), a 5-level scale (from pre-puberty (=1) to post-puberty (=5)), which was converted to a 4-level scale, combining those in late and post puberty into a single category, given that very few participants (<0.1%) were in post-puberty. Number of hours per week spent on screens for purposes other than homework was also estimated from the youth screen time survey, to investigate relationships between anxiety and screen time. All models including fMRI parameters were further adjusted for time of scan (Hu et al., 2023) and percent of frames censored for motion for the analyzed run. At each level of analysis, model results were adjusted for the False Discovery Rate (FDR) (Benjamini and Hochberg, 1995). At the network level, the FDR correction was applied across topological properties of each network. At the regional level, for each topological property the FDR correction was applied across regions of a particular network, assuming that p-values would likely be correlated. For each anatomical region, the correction was applied across morphometric properties. The significance level was set at $\alpha = 0.05$. All data analyses were performed in the Harvard Medical School High Performance Cluster Computing, using the software MATLAB (release 2023b, Mathworks, Inc).

3. RESULTS

Participant demographic characteristics are provided in Table 1. The race and ethnicity distribution of the sample (about 65% White and almost 80% non-Hispanic) reflected the distribution of these demographics of the larger ABCD cohort. Median (IQR) anxiety t-scores were 51 (2), and 144 (3.2%) youth had GAD, 122 (3.5%) had SAD and 150

(4.3%) had separation anxiety. Three-hundred-ninety (11%) had at least subclinical anxiety or a diagnosis of a related disorder. Older youth, who spent more time on electronic devices, and/or with mixed race had higher anxiety ($\beta = 0.04$ to 1.15 , $95\%CI = [0.01, 2.18]$, $P \leq 0.03$). Black youth and those from higher income families had lower anxiety ($\beta = -1.44$ to -0.05 , $95\%CI = [-2.51, -0.01]$, $P < 0.05$). Non-Hispanic youth had lower odds of GAD (aOR = 0.36 , $95\%CI = [0.16, 0.79]$, $P = 0.01$). Females had higher odds of having any anxiety compared to males (aOR = 1.39 , $95\%CI = [1.03, 1.89]$, $P = 0.04$). Additional associations and model statistics are provided in Table 2.

3.1 Associations between anxiety and brain topological properties

At the scale of individual regions, higher local connectedness (degree) and topological centrality of the cerebellum and higher amygdala degree were associated with higher anxiety ($\beta = 0.04$ to 0.07 , $95\%CI = [0.01, 0.11]$, $P < 0.05$). Also, higher topological clustering of the hippocampus and lower clustering of the ventral PFC and temporoparietal regions were associated with higher anxiety ($\beta = 0.06$, $95\%CI = [0.02, 0.09]$, $P = 0.01$ for the hippocampus, and $\beta = -0.07$, $95\%CI = [-0.92, -0.02]$, $P < 0.05$ for PCF). Furthermore, higher centrality of the parahippocampal cortex, hippocampus, and cerebellum were associated with higher odds of GAD (aOR = 1.20 to 1.32 , $95\%CI = [1.03, 1.54]$, $P \leq 0.04$), whereas higher clustering in ventral PFC was associated with lower odds of GAD (aOR = 0.69 , $95\%CI = [0.55, 0.87]$, $P = 0.01$). Higher centrality of temporoparietal regions was associated with lower odds of SAD (aOR = 0.74 , $95\%CI = [0.57, 0.95]$, $P = 0.04$). Similarly, higher clustering in superior parietal regions,

somatomotor regions, medial, lateral and ventral PFC, insula, retrosplenial cortex and bilateral cerebellum was associated with lower odds of SAD (aOR = 0.58 to 0.80, 95%CI = [0.55, 0.99], $P < 0.05$). The spatial distribution of these associations are shown in Figure 1.

At the scale of individual networks, higher connectivity between the thalamus and the rest of the brain was associated with higher anxiety ($\beta = 0.06$, 95%CI = [0.02, 0.10], $P = 0.04$), and higher connectivity within the thalamus was associated with higher odds of GAD (aOR = 1.34, 95%CI = [1.12, 1.59], $P = 0.01$; Figure 2). Higher robustness, efficiency, clustering and topological stability of the temporoparietal network were associated with lower odds of SAD (aOR = 0.72 to 0.79, 95%CI = [0.59, 0.95], $P \leq 0.03$), and similarly for the frontoparietal network (aOR = 0.77 to 0.79, 95%CI = [0.62, 0.97], $P < 0.04$; Figure 3). Also, higher topological robustness, stability and efficiency of the cerebellum was associated with lower odds of SAD (aOR = 0.73 to 0.79, 95%CI = [0.58, 0.99], $P < 0.05$). In contrast, Finally, higher connectivity of the hippocampus was associated with higher odds of SDA (aOR = 1.42, 95%CI = [1.16, 1.73], $P = 0.01$). Model statistics are provided in Table 3. Additional associations based on analyses of the second rs-fMRI run are provided in Table S1.

3.2. Associations between anxiety and brain network controllability

At the scale of individual regions, higher feedback costs in bilateral basal ganglia were associated with higher anxiety ($\beta = 0.04$ to 0.05, 95%CI = [0.01, 0.09], $P < 0.05$), and

higher costs in the parahippocampal cortex, insula, and the cerebellum were associated with higher odds of GAD (aOR = 1.17 to 1.21, 95%CI = [1.05, 1.37], $P \leq 0.03$). At the network scale, median control costs across the network comprising the cerebellum and basal ganglia and similarly for the networks comprising the cerebellum, basal ganglia and thalamus, and cerebellum, basal ganglia, thalamus, and amygdala, respectively were associated with higher anxiety ($\beta = 0.04$ to 0.05 , 95%CI = [0.01, 0.10], $P \leq 0.04$). In addition, higher median control costs in the network including the cerebellum, basal ganglia and thalamus were associated with higher odds of GAD (aOR = 1.16, 95%CI = [1.01, 1.34], $P = 0.04$). Model statistics are provided in Table 3. Associations between the anxiety and the topological centrality of the cerebellum and costs of regulating the network that includes this structure are shown in Figure 4.

3.3 Associations between anxiety and intrinsic dynamics

Higher signal fluctuation amplitude in the medial and dorsal PFC was associated with lower anxiety ($\beta = -0.05$ to -0.04 , 95%CI = [-0.09, -0.01], $P \leq 0.04$). Similarly, higher fluctuation amplitude in the medial and dorsal PFC, and frontomedial regions of the salience network was associated with lower odds of GAD (aOR = 0.61 to 0.71, 95%CI = [0.49, 0.88], $P \leq 0.02$). Higher temporal variability of regional modularity (clustering) of the right basal ganglia was associated with higher odds of GAD (aOR = 1.28, 95%CI = [1.06, 1.55], $P = 0.02$). In contrast, higher temporal variability of regional connectedness (degree) of the left hippocampus, right cerebellum, bilateral retrosplenial and posterior cingulate cortex and left temporal and PFC regions, and higher centrality variability of

the left hippocampus was associated with lower odds of SAD (aOR = 0.73 to 0.84, 95%CI = [0.62, 0.96], $P \leq 0.03$).

3.4 Associations between anxiety and structural brain properties

White matter hyperintensity in the left temporal pole and the right superior temporal gyrus was associated with higher anxiety ($\beta = 0.05$ to 0.06 , 95%CI = [0.01, 0.11], $P \leq 0.03$). Higher gray matter volume of the left lingual gyrus was associated with lower odds of GAD (aOR = 0.74, 95%CI = [0.59, 0.93], $P = 0.03$), whereas higher volume of the superior temporal gyrus and left putamen was associated with higher odds of SAD (aOR = 1.28 to 1.44, 95%CI = [1.03, 1.78], $P \leq 0.02$). Higher cortical thickness of the left pars opercularis, left superior temporal gyrus and right precentral gyrus was associated with higher odds of SAD (aOR = 1.30 to 1.40, 95%CI = [1.06, 1.74], $P \leq 0.03$). Associations between higher likelihood of SAD and cortical thickness of superior temporal gyrus and pars opercularis, and opposite relationships between the efficiency of the networks to which these regions belong are shown in Figure 3. Finally, higher cortical thickness of the left lateral orbitofrontal cortex was associated with higher odds of any anxiety (aOR = 1.19, 95%CI = [1.05, 1.35], $P = 0.02$). Model statistics and additional associations are provided in Tables 4 and S2.

In brain regions where both structural and functional properties were altered in youth with anxiety, relationships between morphometric and topological characteristics were further examined. Higher gray matter volume of the right temporal pole was associated

with higher global clustering in the right limbic network ($\beta = 0.05$ to 0.06 , $95\%CI = [0.01, 0.10]$, $P \leq 0.04$), and lower white matter intensity in this region was associated with lower fragility in the network ($\beta = -0.06$, $95\%CI = [-0.10, -0.02]$, $P = 0.02$). Higher cortical thickness of the right superior temporal gyrus was associated with lower robustness, efficiency and topological stability of the temporoparietal network ($\beta = -0.05$ to -0.04 , $95\%CI = [-0.08, -0.01]$, $P \leq 0.03$). Higher gray matter volume and white matter intensity of the left pars opercularis was associated with higher median connectivity, robustness and topological stability of the left ventral prefrontal network ($\beta = 0.04$ to 0.15 , $95\%CI = [0.01, 0.21]$, $P \leq 0.04$). Higher white matter intensity of the left superior temporal gyrus was associated with higher connectivity, robustness, stability and global clustering ($\beta = 0.08$ to 0.15 , $95\%CI = [0.03, 0.20]$, $P \leq 0.01$) in parts of the left default mode network, and lower fragility of the parts of the left social and limbic networks overlapping with this region ($\beta = -0.06$ to -0.04 , $95\%CI = [-0.12, -0.01]$, $P < 0.05$) in the left social and limbic network. Higher white matter intensity of the right precentral gyrus was associated with higher median connectivity of parts of the right dorsal attention network ($\beta = 0.09$ to 0.10 , $95\%CI = [0.04, 0.16]$, $P \leq 0.03$) and higher cortical thickness was associated with lower fragility in this network ($\beta = -0.06$, $95\%CI = [-0.09, -0.02]$).

3.6 Associations between anxiety, brain characteristics and cognitive outcomes

Higher anxiety was directly associated with lower scores in the Little Man Task ($\beta = -0.05$, $95\%CI = [-0.09, -0.01]$, $P = 0.02$), and no other cognitive outcomes. Performance in this task was positively associated with the topological centrality of regions in the

orbitofrontal, limbic, somatomotor, dorsal attention, and salience networks ($\beta = 0.07$ to 0.71 , $95\%CI = [0.01, 1.08]$, $P \leq 0.04$). These associations were attenuated by higher anxiety (interaction $\beta = -0.75$ to -0.11 , $95\%CI = [-0.98, -0.02]$, $P \leq 0.04$). Similarly, higher local clustering of regions in the somatomotor and salience networks was associated with better task performance ($\beta = 0.38$ to 0.57 , $95\%CI = [0.03, 0.92]$, $P \leq 0.04$), but these associations were attenuated by anxiety (interaction $\beta = -0.60$ to -0.39 , $95\%CI = [-0.95, -0.04]$, $P < 0.05$). Higher connectivity within the bilateral somatomotor network and lower connectivity within the right social network were associated with better task performance ($\beta = 0.48$ to 0.55 , $95\%CI = [0.11, 0.92]$, $P \leq 0.02$, and $\beta = -0.57$, $95\%CI = [-0.95, -0.19]$, $P = 0.02$, respectively). These relationships were adversely moderated by anxiety (interaction $\beta = -2.09$ to -1.78 , $95\%CI = [-3.45, -0.42]$, $P \leq 0.02$, and $\beta = 3.01$, $95\%CI = [1.02, 4.99]$, $P = 0.01$, respectively). Lower segregation (modularity) of the bilateral somatomotor network, salience and social networks and left basal ganglia was associated with better performance ($\beta = -0.49$ to -0.12 , $95\%CI = [-0.89, -0.09]$, $P \leq 0.03$), but these relationships were also inversely moderated by anxiety (interaction $\beta = 0.49$ to 0.57 , $95\%CI = [0.14, 0.95]$, $P \leq 0.03$). Lower fragility of the somatomotor and basal ganglia networks was associated with better task performance ($\beta = -0.39$ to -0.19 , $95\%CI = [-0.88, -0.05]$, $P \leq 0.02$), and anxiety inversely moderated this relationship (interaction $\beta = 0.59$ to 0.45 , $95\%CI = [0.08, 0.98]$, $P \leq 0.02$). Finally, higher feedback costs associated with the insula's control were

associated with better performance $\beta = 0.64$, 95%CI = [0.23, 1.04], $P = 0.01$, but this relationship was attenuated by anxiety ($\beta = -1.07$, 95%CI = [-1.74, -0.40], $P = 0.01$). Model statistics are provided in Table 5. Additional associations between brain topological properties and performance in this task or other tasks were not associated with or moderated by anxiety.

4. DISCUSSION

Adolescence is a period of extensive biological and social changes that have profound positive or negative impact on brain maturation. During this period, endogenous and exogenous risk factors and adverse experiences can have detrimental effects on brain circuit rewiring, leading to poor cognitive and mental health outcomes. However, relationships between brain structure, circuit wiring, mental health and cognitive function during this period are incompletely understood. Focusing on anxiety (and related disorders), which is relatively common in adolescents, this cross-sectional study investigated its relationships with structural, functional and dynamic brain characteristics, as well as its direct and indirect (through the brain) relationships with cognitive function across domains. To robustly characterize these relationships, it analyzed neuroimaging, neurocognitive and anxiety data from almost 3,500 youth from the ABCD study.

Structural and functional brain characteristics of distributed brain networks and regions were associated with anxiety and/or GAD and SAD. These included the cerebellum,

basal ganglia, thalamus, amygdala and regions of the frontoparietal and temporoparietal networks, such as the superior temporal gyrus, the temporal pole, the insula and the pars opercularis. Several of these regions are brain hubs that integrate multidomain information in response to cognitive demands across domains, and undergo significant topological changes in adolescence (van den Heuvel and Sporns, 2013; Oldham and Fornito, 2018; Bagarinao et al, 2020). Although our study was cross-sectional, and thus directional relationships between anxiety and brain alterations could not be established, our findings suggest that aberrant organization of distributed circuits and morphometric alterations of structures that play key roles in cognitive processing may constitute the neural substrates of anxiety and be associated with higher likelihood of related disorders.

A growing number of studies have reported associations between anxiety and alterations in the cerebellum's structural characteristics, topological organization and abnormal connections to other regions (Talati et al, 2013; Lee et al, 2021; Chin and Augustine, 2025). The cerebellum undergoes significant structural changes in adolescence (its volume has an inverted U-shaped trajectory with a peak around age ~12 years, its subdivisions mature at distinct rates (Tiemeier et al, 2009), and it completes its development later than many other brain structures (Gaiser et al 2024). In addition, its widespread long-range connections with the rest of the brain also undergo accelerated maturation in adolescence. In particular, the cerebellum is part of multiple networks that develop significantly in order to support high-level cognitive processes,

including the cerebellum-cingulo-operculum network (Clark et al, 2022), the limbic network (Jung et al, 2022), a key network that connects the cerebellum to the basal ganglia through the thalamus and supports affective processing (Bostan and Strick, 2018; Pierce and Peron, 2020), and circuits connecting the cerebellum to the prefrontal cortex, the frontoparietal, default-mode and temporoparietal networks, and structures such as the superior temporal and lingual gyri (O'Reilly et al, 2010; Macher et al, 2014; Igelstrom et al, 2017; Stockert et al, 2021; Cerevolo et al, 2021). Through these extensive connections, the cerebellum plays a key role in emotional processing and regulation and social function, both of which are impacted by anxiety in adolescents (Young et al, 2019;. Our findings, including associations between topological properties of the cerebellum and anxiety, and morphometric alterations of structures connected to it, are in agreement with prior work on relationships between the cerebellum and anxiety, and provide mechanistic insights on neural correlates of emotional and social function in youth with anxiety, including disorders such as SAD. In addition, we show that the cost of controlling the dynamics of cerebellum-thalamus-basal ganglia network is higher in youth with anxiety, potentially as a result of aberrant connectivity and topological organization of the cerebellum, but also the thalamus and the basal ganglia, both of which play key roles in emotion regulation (Haber and Calzavara, 2009; Marchand, 2010). Higher connectivity of the thalamus and stronger connections with the rest of the brain was associated with anxiety, including GAD. Higher volume of the putamen was associated with higher likelihood of SAD, whereas higher efficiency and robustness of connections between the basal ganglia and the prefrontal cortex were

associated with lower likelihood of SAD. These findings are in agreement with prior studies that have reported topological and structural anomalies in circuits including both structures (Xie et al, 2021).

Anxiety was also associated with morphometric and topological alterations of the frontoparietal and temporoparietal networks and their constituent structures. Higher efficiency, topological robustness and stability of these networks were associated with lower likelihood of SAD. In contrast, higher fragility and topological segregation of the PFC, and higher cortical thickness of the pars opercularis were associated with higher likelihood of the disorder. The frontoparietal control network plays a key regulatory role in mental health, and anxiety has been associated with alterations in its organization and lower functioning of the network (Sylvester et al, 2012; Cole et al, 2014; Kenwood et al, 2021; Baggio et al, 2025a, 2025b). It undergoes significant reorganization in adolescence, to support cognitive control, and is thus vulnerable to miswiring (Luna et al, 2010; Di Martino et al, 2010), which can increase the risk of mental health disorders. Similarly, the temporoparietal network, which plays a key role in social cognition, matures significantly in adolescence (Blakemore 2008, 2012; Mills et al 2012; Solle-Padules et al, 2016), and is vulnerable to stressors. Prior studies have reported associations between structural and connectivity alterations in this network and anxiety, especially SAD (Tei et al, 2020; Swaltha et al, 2021; Chang and Englemann, 2024). In our sample, higher efficiency, robustness and topological stability of the temporoparietal network was associated with lower likelihood of SAD, whereas hyperintensity of the

temporal pole, higher thickness and gray matter volume of the superior temporal gyrus was associated with higher anxiety and/or likelihood of SAD. In addition, higher fragility and modularity of the social network (which overlaps with elements of the temporoparietal network and includes the superior temporal gyrus) were associated with higher likelihood of GAD and/or SAD. Together, these findings suggest that anxiety may impact widespread cortical and subcortical resting-state networks, which constitute the backbone of the brain's neural circuitry, and play a ubiquitous role in cognitive function.

To elucidate the impact of brain alterations in youth with anxiety on cognitive function, we examined direct and indirect (through the brain) associations between anxiety and cognitive function, leveraging the extensive neurocognitive assessments of the ABCD study. In agreement with prior findings, anxiety was specifically associated with worse performance (lower response accuracy) in the Little Man Task, which measures visuospatial attention and spatial reasoning, and mental rotation skills. Prior studies have reported deficits in visuospatial processing and mental rotation in individuals with anxiety (Borst et al, 2012; Ramirez et al; Thoresen et al, 2016; Kalanthroff, 2024), which may be more pronounced in girls (Lennon-Maslin and Quaiser-Pohl, 2024). Anxiety was also indirectly associated with performance in the Little Man Task, through its interaction with the brain. Specifically, it attenuated positive associations between topological properties of the salience and the somatomotor network and response accuracy. Both networks play important roles in visuospatial processing and mental rotation (Jordan et al, 2001; Windischberger et al, 2003; Uddin, 2016; Toller et al, 2018; Tosoni et al,

2023). Anxiety also inversely moderated negative associations between topological fragility and modularity of the basal ganglia and task performance, another structure involved in visuospatial processing (DeLong, 2000), and negative associations between connectivity and modularity of the social network and task performance. Some of the structures and regions that form the social network are involved in visuospatial processing (Blanke et al, 2005; Schenden and Stern, 2007; Anomal et al, 2020). No other direct or indirect relationships between cognitive and anxiety were identified, which may in part be due to the relatively low prevalence of anxiety in our sample (less than 5% with any anxiety disorder, and on average anxiety scores were low).

Together, our findings suggest complex relationships between adolescent anxiety, topological and structural brain properties of interconnected regions (including brain hubs where multidomain information is integrated), and cognitive processing. These findings are in agreement with prior work but also provide a number of novel insights into the neural correlates of anxiety, especially the cerebellum-thalamus-basal ganglia circuit, which supports multidomain cognitive function, plays a key regulatory role in emotion processing, and its constituent structures mature at distinct paces during adolescence. In addition to topological and morphometric alterations of this network, its dynamic controllability was also impacted in youth with anxiety. In prior work, we have shown that despite its extensive connections to other brain regions, the cost associated with controlling regions of this network is low (Lim et al, 2025), likely due to key roles of these structures in sensorimotor function that requires them to be easily regulated by

endogenous processes. Although youth with anxiety had aberrantly higher connectedness (and topological centrality) in these regions, and stronger connections with the rest of the brain, topological alterations may not be the only correlate of higher control costs. This network plays an important regulatory role in emotional processing, which is adversely impacted by anxiety. Our findings provide mechanistic evidence that the dysregulation of emotions in youth with anxiety is associated with difficulty in controlling the dynamics of its underlying neural substrates. Finally, we also found that the organization of the frontoparietal and temporoparietal networks, especially higher efficiency and topological resilience, was associated with lower likelihood of SAD, which suggests that specific aspects of these networks' wiring may be protective for mental health, in agreement with prior work, including ours (Hu and Stamoulis, 2024).

Despite its strengths, including a unique sample of almost 3,500 sample that captures the heterogeneity of the adolescent brain, a comprehensive investigation of neural and cognitive correlates of anxiety, including brain circuit organization, morphometric properties, intrinsic dynamics and their controllability (using a highly novel to the field approach), and an extensive set of cognitive tasks, this study also had some limitations. First, the prevalence of anxiety in the sample was relatively low, although the sample had sufficient statistical power to identify widespread neural correlates of anxiety. Second, like any retrospective investigation, the study was limited by the decisions of the ABCD investigators, especially in terms of neurocognitive assessments. Nevertheless, the ABCD includes extensive cognitive tasks that measure domains that

are adversely impacted by anxiety, although additional tasks on emotion processing were not available. Finally, bidirectional associations between anxiety and brain alterations could not be established. However, the ABCD study is longitudinal and additional assessments have become available. Thus, a future investigation could track the developmental trajectories of youth with anxiety over time and establish such relationships.

5. CONCLUSIONS

This study makes an important scientific contribution to the field's incomplete understanding of the neural substrates of mental health problems in adolescence, a period of increased risk of neural miswiring and psychopathologies. It also provides robust mechanistic insights into brain function in youth with anxiety and impacts of their structural and topological alterations on cognitive function, as well as moderations of brain-cognition relationships by anxiety. It specifically identified alterations in brain circuits and structures that play key regulatory roles in emotional processing, social function and decision-making, all of which evolve significantly in adolescence. Identified structures and circuits could become targets of precise intervention, to improve cognitive and mental health outcomes in this population.

Declarations of interest: None

Data and code availability: The data that support the findings of this study are publicly available at the National Institute of Mental Health Data Repository: <https://www.nbdc-datahub.org/>

The assession number (DOI) associated with study in NDA repository is:

<http://dx.doi.org/10.15154/1523041>

Codes used in the data analyses are available at:

- <https://github.com/cstamoulis1/Next-Generation-Neural-Data-Analysis-NGNDA/>

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Madhumitha Manjunath: Data curation, Formal analysis, Software, Visualization, Validation, Writing (original draft and review and editing)

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FIGURES

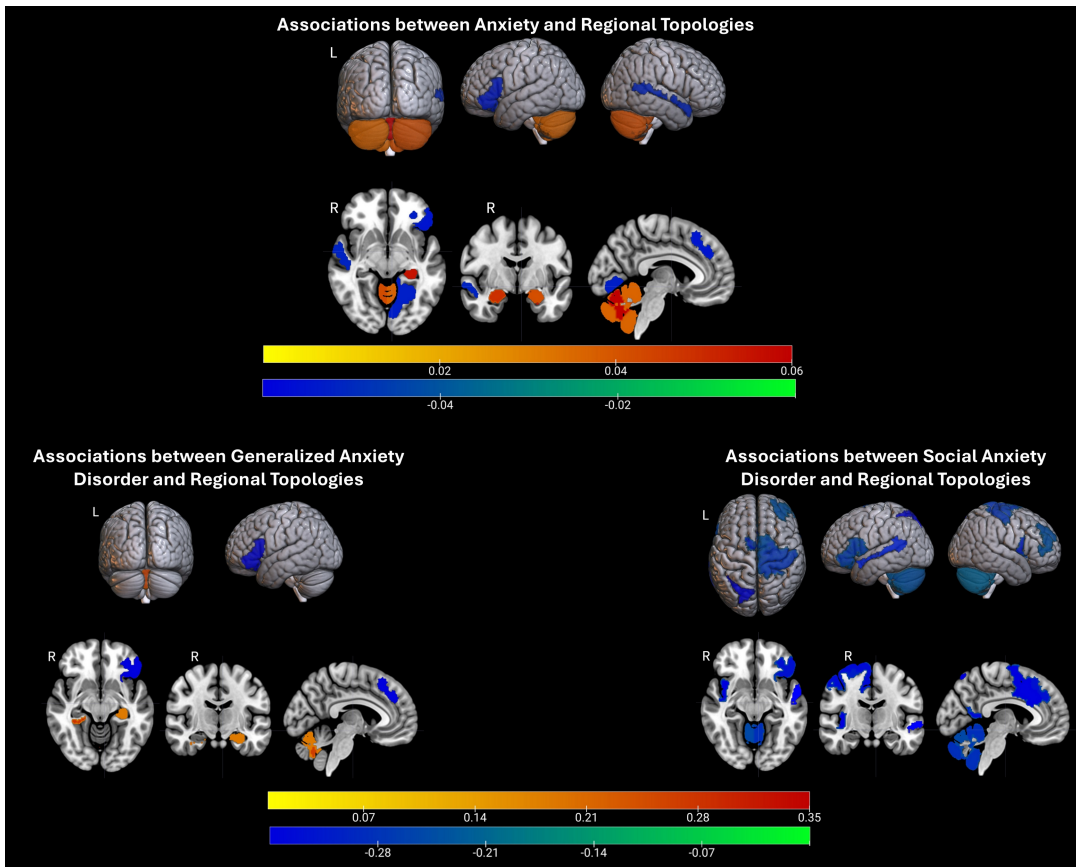


Figure 1: Top panel: Positive associations (yellow to red colors) between anxiety (t-scores) and regional connectedness and topological centrality (regional importance in the network), and negative associations (blue to green colors) between anxiety and regional segregation (clustering). Bottom left panel: Positive associations between Generalized Anxiety Disorder and regional centrality, and negative associations with regional segregation. Bottom right panel: Negative associations between Social Anxiety Disorder and both topological centrality and regional segregation.

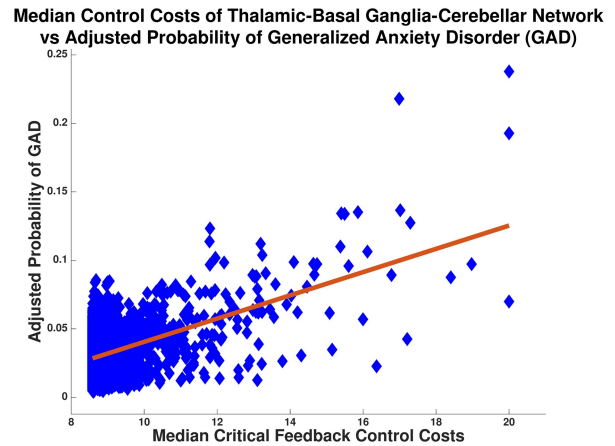
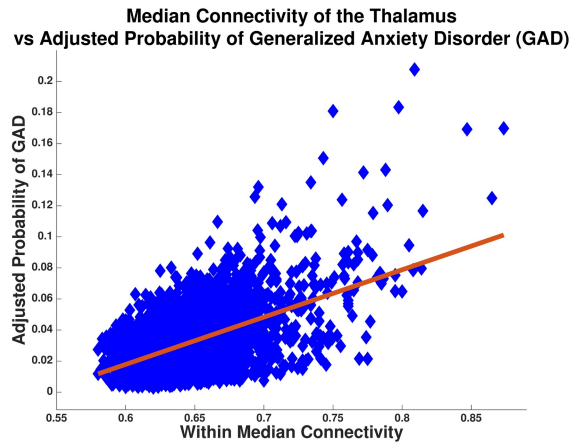


Figure 2: Left panel: Association between median connectivity of the thalamic network and probability of Generalized Anxiety Disorder (GAD; left panel). Right panel: Association between feedback costs associated with the controllability of the thalamic-basal ganglia-cerebellar network and probability of GAD. Probability estimates are based on logistic regression models with adjustments for demographic and imaging parameters.

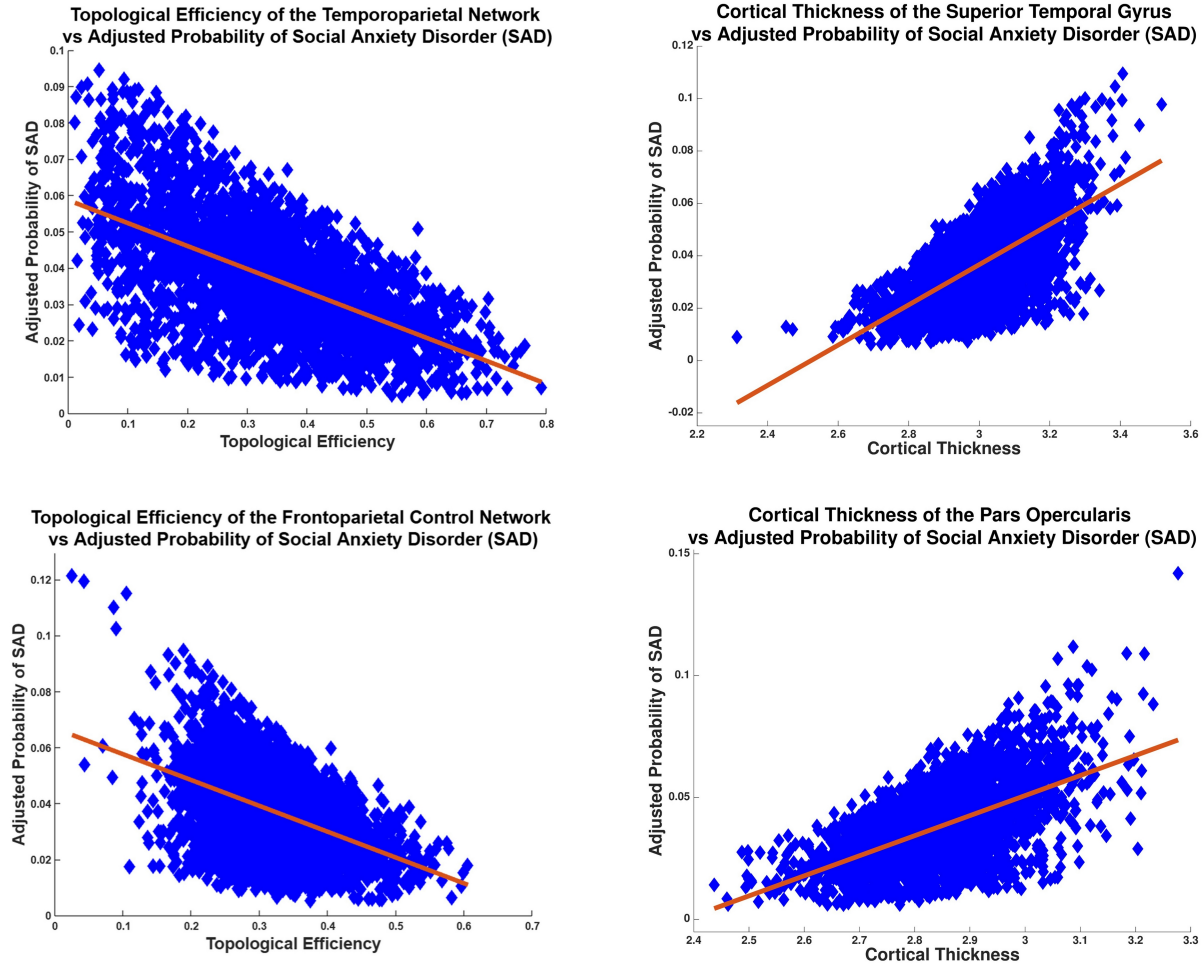


Figure 3: Left top and bottom panels: Negative associations between the topological efficiency of the temporoparietal (top) and frontoparietal (bottom) networks and probability of Social Anxiety Disorder (SAD). Right top panel: Positive association between thickness of the superior temporal gyrus and probability of SAD. Right bottom panel: Positive association between thickness of the pars opercularis and probability of SAD. Probability estimates are based on logistic regression models with adjustments for demographic and imaging parameters.

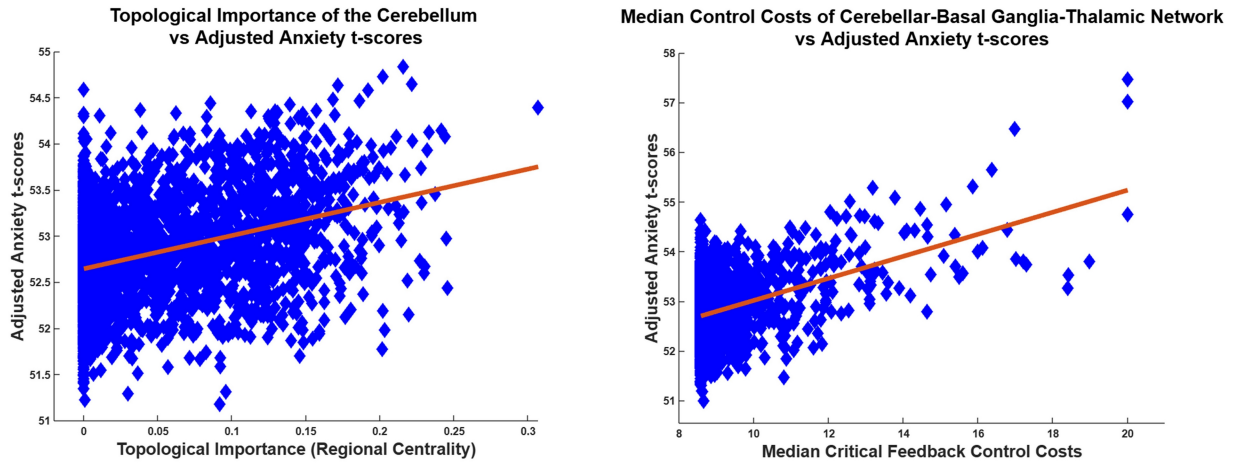


Figure 4: Positive associations between the topological centrality of the cerebellum and anxiety (left panel), and positive associations between cost of controlling this network and anxiety (right panel).

TABLES

Table1. Participant characteristics (N = 3499)

Characteristics	Measure/Categories	Statistic
		Median (IQR)
Age (Years)		12.0 (1.08)
BMI Z-Score		-0.28 (1.17)
Screen time (Hours/Week)		20.5 (18.13)
		N (%)
Sex	Male	1715 (49.01)
	Female	1782 (50.93)
	Missing	2 (0.06)
Race	White	2244 (64.31)
	Black	529 (15.12)
	Asian	77 (2.20)
	Mixed Race	414 (11.73)
	Other	183 (5.23)
	Missing	52 (1.41)
Hispanic	Hispanic	768 (21.95)
	Non-Hispanic	2696 (77.05)
	Missing	35 (1)
Family Income (\$)	< 25,000	344 (9.83)
	25,000 to 49,999	416 (11.89)

	50,000 to 74,999	425 (12.15)
	75,000 to 99,999	450 (12.86)
	100,000 to 199,999	1091 (31.18)
	$\geq 200,000$	508 (14.52)
	Missing	265 (7.57)
Pubertal Stage	Prepuberty	633 (18.09)
	Early puberty	749 (21.41)
	Mid-puberty	1130 (32.29)
	Late/Postpuberty	801 (22.89)
	Missing	186 (5.32)

Table 2. Statistics of mixed effects models testing associations between demographics parameters with anxiety scores, individual anxiety disorders or any anxiety. *CI: confidence interval; aOR: adjusted odds ratio.

ANXIETY T-SCORES			
Parameters	Beta	95% CI*	P-value
Age	0.062	[0.023, 0.101]	0.002
Race: Black*	-1.440	[-2.51, -0.367]	0.009
Race: Mixed*	1.151	[0.122, 2.179]	0.028
Family income	-0.050	[-0.099, -0.001]	0.045
Screentime	0.044	[0.007, 0.080]	0.021
GENERALIZED ANXIETY DISORDER			
Parameters	aOR	95% CI*	P-value
Ethnicity: Hispanic*	0.357	[0.161,0.795]	0.012
ANY ANXIETY			
Parameters	aOR	95% CI*	P-value
Family income	0.929	[0.865, 0.997]	0.041
Race: Mixed*	2.442	[1.228, 4.859]	0.011
Sex	1.395	[1.029, 1.889]	0.031

Table 3. Statistics of mixed effects models testing associations between anxiety and network topological properties, fluctuation amplitude and median feedback control costs, based on the best-quality resting-state fMRI run. All reported p-values have been adjusted for the False Discovery Rate (FDR). *CI: confidence interval.

ANXIETY T-SCORES				
Network	Properties	Beta	95% CI*	P-value
Thalamus	Median Connectivity (out)	0.061	[0.018, 0.104]	0.045
Cerebellar-Basal Ganglia (Bilateral)	Median Feedback Control Costs	0.045 to 0.046	[0.007, 0.083]	≤ 0.030
Cerebellar-Basal Ganglia-Thalamic (L)	Median Feedback Control Costs	0.046	[0.004, 0.089]	0.031
Cerebellar-Basal Ganglia-Amygdalo-Thalamic (Bilateral)	Median Feedback Control Costs	0.045 to 0.050	[0.002, 0.097]	≤ 0.039
GENERALIZED ANXIETY DISORDER*				
Network	Properties	aOR	95% CI*	P-value
Thalamus	Median Connectivity (in)	1.337	[1.124, 1.591]	0.009
Cerebellar-Basal Ganglia-Thalamic (L)	Median Feedback Control Costs	1.160	[1.007, 1.337]	0.039
SOCIAL ANXIETY DISORDER*				
Structure	Properties	aOR	95% CI*	P-value
Frontoparietal Control (L)	Robustness	0.771	[0.619, 0.961]	0.037
	Efficiency	0.787	[0.642, 0.964]	0.037
	Topological stability	0.773	[0.621, 0.965]	0.036
	Global Cluster Coefficient	0.791	[0.649, 0.963]	0.037

	Modularity	1.301	[1.067, 1.588]	0.037
Temporoparietal (L)	Robustness	0.723	[0.632,0.908]	0.0301
	Efficiency	0.772	[0.632, 0.943]	0.0301
	Global Clustering Coefficient	0.788	[0.649, 0.955]	0.032
	Topological stability	0.745	[0.597, 0.929]	0.0301
Hippocampus (R)	Median Connectivity (in)	1.416	[1.157, 1.734]	0.006
Cerebellum (bilateral)	Network Robustness	0.728 to 0.767	[0.575, 0.946]	≤ 0.036
	Modularity	1.324 to 1.337	[1.116, 1.586]	≤ 0.011
	Topological stability	0.728 to 0.739	[0.575, 0.911]	≤ 0.036
Cerebellum (R)	Efficiency	0.798	[0.651, 0.977]	0.046
	Global Cluster Coefficient	0.763	[0.632, 0.922]	0.014

Table 4. Statistics of mixed effects models testing associations between anxiety and morphometric brain properties. All reported p-values have been adjusted for the False Discovery Rate (FDR). *CI: confidence interval; aOR: adjusted odds ratio.

ANXIETY T-SCORES				
Structure	Properties	Beta	95% CI*	P-value
Temporal pole (L)	White matter intensity	0.052	[0.012, 0.093]	0.035
GENERALIZED ANXIETY DISORDER				
Structure	Properties	aOR	95% CI*	P-value
Lingual gyrus (L)	Gray matter volume	0.738	[0.588, 0.926]	0.027
SOCIAL ANXIETY DISORDER				
Structure	Properties	aOR	95% CI*	P-value
Pars opercularis (L)	Cortical thickness	1.333	[1.085, 1.636]	0.012
Superior temporal gyrus (L)	Cortical thickness	1.326	[1.069, 1.645]	0.015
	Gray matter volume	1.438	[1.160, 1.782]	0.003
Precentral gyrus (R)	Cortical thickness	1.365	[1.071, 1.739]	0.035
Putamen (L)	Gray matter volume	1.277	[1.032, 1.579]	0.025
ANY ANXIETY				
Structure	Properties	aOR	95% CI*	P-value
Lateral orbitofrontal cortex (L)	Cortical thickness	1.190	[1.048, 1.352]	0.022

Table 5. Statistics of mixed effects moderation models testing associations between brain measures and number of correct responses in the Little Man task, and interactions between anxiety and brain measures. All reported p-values have been adjusted for the False Discovery Rate (FDR). *CI: confidence interval.

Little Man Task (Sum of Correct Responses)				
Network	Parameters	Beta	95% CI*	P-value
Somatomotor (Bilateral)	Median Connectivity (in)	0.481 to 0.556	[0.113, 0.917]	< 0.020
	Interaction: anxiety t-score x Median Connectivity (in)	-2.095 to -1.777	[-3.445, -0.423]	≤ 0.014
	Modularity	-0.495 to -0.493	[-0.857, -0.150]	≤ 0.015
	Interaction: anxiety t-score x Modularity	0.522 to 0.524	[0.147, 0.897]	≤ 0.012
Somatomotor (L)	Fragility	-0.390	[-0.723, -0.057]	0.024
	Interaction: anxiety t-score x Fragility	0.448	[0.078, 0.817]	0.020
Salience (Bilateral)	Modularity	-0.447 to -0.466	[-0.804, -0.092]	≤ 0.036
	Interaction: anxiety t-score x Modularity	0.468 to 0.511	[0.104, 0.857]	≤ 0.032
Social (Bilateral)	Modularity	-0.456 to -0.538	[-0.896, -0.118]	≤ 0.032
	Interaction: anxiety t-score x Modularity	0.521 to 0.571	[0.161, 0.954]	≤ 0.020
Social (R)	Median Connectivity (in)	-0.572	[-0.950, -0.194]	0.012

	Interaction: anxiety t-score x Median Connectivity (in)	3.009	[1.021, 4.997]	0.014
Basal Ganglia (L)	Modularity	-0.124	[-0.817, -0.471]	0.013
	Interaction: anxiety t-score x Modularity	0.489	[0.140, 0.838]	0.009
	Fragility	-0.194	[-0.881, -0.537]	0.010
	Interaction: anxiety t-score x Fragility	0.595	[0.209, 0.981]	0.007
Salience (Insula)l (R)	Feedback Control Costs	0.637	[0.230, 1.044]	0.007
	Interaction: anxiety t-score x Feedback Control Costs	-1.073	[-1.745, -0.401]	0.006

SUPPLEMENTAL MATERIALS

Table S1. Statistics of mixed effects models testing associations between anxiety (t-scores and individual anxiety disorders) and topological properties and median feedback control costs, estimated from the second fMRI run. All reported p-values have been adjusted for the False Discovery Rate (FDR). *CI: confidence interval.

ANXIETY T-SCORES (DSM-5)				
Network	Properties	Beta	95% CI*	P-value
Thalamus (L)	Modularity	0.268	[0.084, 0.451]	0.037
Temporoparietal (L)		0.065	[0.028, 0.0108]	0.011
Generalized Anxiety Disorder*				
Network	Properties	aOR	95% CI*	P-value
Temporoparietl (L)	Modularity	1.476	[1.168, 1.865]	0.001
	Fragility	1.309	[1.051,1.631]	0.041
	Robustness	0.717	[0.539, 0.952]	0.041
	Efficiency	0.749	[0.582, 0.966]	0.041
	Topological Stability	0.722	[0.546, 0.954]	0.041
Reward (L)	Fragility	1.367	[1.142, 1.636]	0.005
Social (L)	Fragility	1.390	[1.125, 1.718]	0.018
Social Anxiety Disorder*				
Network	Properties	aOR	95% CI*	P-value
Salience (L)	Robustness	0.763	[0.598, 0.974]	0.046
	Modularity	1.259	[1.017, 1.559]	0.046
	Median Connectivity (out)	0.706	[0.526, 0.930]	0.036

	Efficiency	0.720	[0.571, 0.908]	0.022
	Global Clustering Coefficient	0.719	[0.575, 0.900]	0.022
	Topological Stability	0.760	[0.595, 0.972]	0.046
Reward (L)	Robustness	0.706	[0.544, 0.917]	0.021
	Fragility	1.226	[1.027, 1.464]	0.033
	Modularity	1.359	[1.085, 1.704]	0.021
	Topological Stability	0.710	[0.547, 0.923]	0.022
	Efficiency	0.721	[0.565, 0.919]	0.021
	Global Clustering Coefficient	0.761	[0.601, 0.965]	0.033
Social (L)	Modularity	1.436	[1.138, 1.812]	0.019
	Topological Stability	0.709	[0.544, 0.925]	0.026
	Robustness	0.715	[0.548, 0.932]	0.026
	Efficiency	0.720	[0.570, 0.910]	0.024
PFC (L)	Robustness	0.734	[0.569, 0.945]	≤ 0.036
	Fragility	1.265	[1.059, 1.510]	
	Modularity	1.302	[1.045, 1.622]	
	Topological Stability	0.737	[0.571, 0.952]	
	Efficiency	0.738	[0.584, 0.932]	
	Global Clustering Coefficient	0.767	[0.607, 0.969]	
Frontal-Thalamic (L)	Robustness	0.722	[0.557, 0.935]	≤ 0.031
	Modularity	1.368	[1.095, 1.709]	
	Topological Stability	0.724	[0.558, 0.940]	
	Efficiency	0.733	[0.578, 0.930]	
Frontol-Basal Ganglia-Thalamic (L)	Robustness	0.740	[0.572, 0.958]	≤ 0.047
	Modularity	1.364	[1.089, 1.709]	
	Topological Stability	0.739	[0.569, 0.959]	

	Efficiency	0.749	[0.591, 0.949]	
Fronto-Amygdalar (L)	Robustness	0.734	[0.568, 0.949]	≤ 0.041
	Efficiency	0.755	[0.596, 0.957]	
	Modularity	1.359	[1.088, 1.697]	
	Topological Stability	0.735	[0.567, 0.953]	
Fronto-Striatal-Thalamic (L)	Robustness	0.742	[0.574, 0.959]	≤ 0.047
	Efficiency	0.748	[0.590, 0.947]	
	Modularity	1.364	[1.088, 1.709]	
	Topological Stability	0.741	[0.571, 0.961]	

Table S2. Statistics of mixed effects models testing associations between anxiety (raw scores) and intrinsic dynamic and morphometrics properties. All reported p-values have been adjusted for the False Discovery Rate (FDR). *CI: confidence interval.

ANXIETY RAW SCORES				
INTRINSIC DYNAMIC PROPERTIES				
Network	Properties	Beta	95% CI*	P-value
Default (L; medial PFC)	Fluctuation Amplitude	-0.050	[-0.089, -0.012]	0.035
Default (L; dorsal PFC)		-0.049	[-0.088, -0.012]	0.036
MORPHOMETRIC PROPERTIES				
Structure	Properties	Beta	95% CI*	P-value
Superior temporal gyrus (R)	White matter Intensity	0.064	[0.017, 0.111]	0.024